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SUPERSEDING
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MILITARY SPECIFICATION

ROCKET MOTOR, MANUFACTURING INERT PARTS FOR, GENERAL SPECIFICATION FOR

This specification has been approved by
Naval Ordnance Systems Command, Department of the Navy

1. SCOPE

1.1 This specification covers the requirements for procurement of inert parts for rocket motors. For the purpose of this specification the terms unit, part, component, and inert assembly are used interchangeably.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on the date of invitation for bid or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-I-6868	Inspection Process, Magnetic Particle
MIL-Q-9858	Quality Program Requirements
MIL-I-45208	Inspection System Requirements

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-453	Inspection, Radiographic

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PUBLICATION

Bureau of Naval Weapons

WR 43	Preparation of Quality Assurance Provisions (Including Classification of Characteristics)

(When requesting any of the preceding documents, give the complete title and designation. Copies of this specification and other unclassified specifications, standards, and publications required by suppliers in connection with specific procurement functions should be obtained from the Commanding Officer, Naval Supply Depot (CDS), 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120.)

2.2 Other publications.

United States Of America Standards Institute

Y14.5	Dimensioning and Tolerancing for Engineering Drawings

(Application for copies should be addressed to the United States of America Standards Institute, 10 East 40th Street, New York, New York 10016).

National Bureau of Standards

Handbook H28	Screw Thread Standards for Federal Services

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D.C. 20402.)

3. REQUIREMENTS

3.1 General. The supplier of inert parts shall meet the requirements of this specification and the requirements of applicable drawings and publications.

3.1.1 Material. The materials and items used in the manufacture of inert parts shall conform to the requirements on the drawings and in the specifications and publications listed in section 2. When a document permits the use of alternate material or items, authorization for use of a specific alternate shall be obtained from the procuring activity.

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3.1.2 Manufacturing process. Unless otherwise specified, any process such as welding, forging, machining, cold forming, etc, capable of meeting the end product requirements (hardness, material, dimensions, structural test, interchangeability, etc) is acceptable. Casting of parts is prohibited unless specified on the drawing.

3.2 Data. No data is required by this specification, or by applicable documents referenced in section 2, unless specified in the contract or order (6.2).

3.3 Preproduction sample. Unless otherwise specified in the contract or order (6.2), a preproduction sample of 15 units shall be manufactured using the materials, equipment and procedures proposed for production. The preproduction sample shall be submitted to an activity designated by the procuring activity for inspection as specified in section 4 for the purpose of determining that, prior to starting production, the supplier is capable of providing units that meet the requirements specified herein. After passing all preproduction inspections, no changes in materials, procedures or equipment shall be made without prior written approval of the procuring activity. The supplier shall be responsible for informing the procuring activity of proposed changes.

3.4 Finish. All components shall be free from burrs, contamination, sharp edges and other defects which could result in malfunction of the equipment or be a safety hazard.

3.5 Dimensioning and tolerancing. Dimensioning and tolerancing shall be in accordance with USAS Y14.5 with the following exceptions:

(a) All tolerance zones are at their true geometric position with respect to each other, i.e., the boundary of perfect form at maximum material condition applies to the complete envelope of the part.

(b) The tolerance zones of all diameters shown about a common axis are concentric with that axis.

3.6 Threads. Unless otherwise specified, all threads shall be in accordance with Handbook H28.

3.7 Structural. Whenever structural tests, including hydrotests, are specified on drawings, the unit shall be capable of meeting these requirements for three additional tests performed by the procuring activity during the guarantee period as defined in the contract or order.

3.8 Discontinuities. There shall be no discontinuities. Defects may be repaired as specified in 6.8.

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3.9 Workmanship. Workmanship shall be that required to ensure conformance to the requirements of this specification and applicable documentation and shall be compatible with the degree of precision required in manufacturing component parts. Unless otherwise specified, all corners shall be broken and fillet radii held to .030 ± .015 inch. The workmanship exhibited in the preproduction samples will be evaluated to determine acceptability (6.2(f)). The approved standards of workmanship shall thereby become a minimum requirement for units offered for acceptance.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Acceptance criteria. The supplier is responsible for assuring that all units offered for acceptance meet the requirements of the applicable documentation. As a minimum recorded inspection, the supplier shall maintain inspection records of all the characteristics in the specification and on the drawings classified as Critical or Major in accordance with WR 43. The inspection data shall be traceable to those characteristics classified as Critical or Major by the serializing of each inert rocket motor and component part. Where dimensional control of a characteristic classified as Minor is provided through automatic equipment, jigs, fixtures, or templates, traceability to a specific serial number is not required, but the vendor, in order to verify the quality being produced by the above mentioned equipment, shall include an inspection sampling and recording plan in the quality program made available to the Government under the provisions of MIL-Q-9858 or MIL-I-45208, as applicable. Unless otherwise specified (6.2), sampling shall be in accordance with MIL-STD-105, inspection level II, single sampling plan for normal inspection, and the Acceptable Quality Level (AQL) of table I.

Table I.

ACCEPTABLE QUALITY LEVELS

Classification of Characteristic	AQL
Critical	*
Major	1.0 percent
Minor	2.5 percent

* For inspection lot sizes of 125 units or less, inspect 100 percent of the lot; for lot sizes of 126 to 3200, use a sample of 125 units with acceptance number zero; for lot sizes greater than 3200, sample in accordance with MIL-STD-105 with acceptance number zero.

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4.2.1 Serialization. The procuring activity shall assign a block of serial numbers and an identifying supplier letter for use on the inert assembly. These numbers shall be used in sequence and shall be assigned to the assembled unit after acceptance inspection. The supplier's serial number on component parts shall be recorded to indicate the unit serial number that parts were assembled with. Unused serial numbers shall be returned to the procuring activity upon completion of the contract. Parts shall be marked in a manner that will not make the parts unserviceable.

4.3 Discontinuities. Unless otherwise specified, discontinuities shall be inspected for in accordance with MIL-I-6868. Areas inaccessible or inappropriate to magnetic particle inspection shall be radiographically inspected in accordance with MIL-STD-453 after the procuring activity has been advised of the areas requiring radiographic inspection.

4.4 Preproduction inspection. Fifteen preproduction units, manufactured in accordance with 3.3, shall be submitted to the activity designated by the procuring activity for inspection. Units shall be inspected for conformance to all requirements and tested for functional characteristics including static test firings. Failure of any unit to meet applicable documentation requirements may result in rejection of the preproduction sample or disqualification of the supplier. Final acceptance of the preproduction sample shall be made by the procuring activity. Further production of the unit by the supplier, prior to completion of these tests or approval by the procuring activity, shall be at the supplier's risk unless otherwise specified in the contract or order.

4.5 Production quality conformance inspection. Production quality conformance inspection shall include the verification of all characteristics classified as Critical or Major. Characteristics classified as Critical or Major shall not be verified by using production tooling as a media of inspection.

4.6 Tests.

4.6.1 Structural. Structural tests shall be performed as specified on detail drawings. The load requirements shall be met as specified.

4.6.2 Hydrotest. Hydrotesting shall be performed in accordance with applicable drawings. The specified testing pressure shall be met for the specified time. Pressure gages shall be accurate to one percent of testing pressure. The unit shall be visually and tactually examined during pressure test for any indication of water leakage. Monitoring the pressure gage alone for an indication of low volume leakage is not sufficient.

4.7 Preparation for delivery inspection. Preservation, packaging, and packing shall conform to section 5.

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5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C, as specified (6.2).

5.1.1 Level A.

5.1.1.1 Cleaning, drying, and preservation application. Cleaning, drying, and preservation application shall be as specified on applicable drawings.

5.1.1.2 Unit packaging. When required, loose parts shall be unit packaged as specified on applicable drawings or as required by the contract or order.

5.1.2 Level C. Units shall be preserved and packaged to ensure protection against deterioration or damage in transit from supplier to initial destination.

5.2 Packing. Packing shall be level A or C, as specified (6.2).

5.2.1 Level A. Units shall be packed in shipping containers provided by the procuring activity or as specified in the contract or order.

5.2.2 Level C. Units shall be packed to ensure protection against deterioration or damage in transit from supplier to initial destination.

5.2.3 Palletizing. When required, shipping containers shall be palletized in accordance with applicable drawings or as required by the contract or order.

5.3 Marking. In addition to any special marking required by the contract or order, unit packages and shipping containers shall be marked in accordance with MIL-STD-129.

5.4 Deviation. If rocket motor inert parts and loaded assembly are manufactured by the same supplier, the requirements of this section and the shipping requirements of the inert assembly drawings do not apply.

6. NOTES

6.1 Intended use. This specification is intended for use in procurement of inert parts of rocket motors.

6.2 Ordering data. Procurement documents should specify, but not be limited to, the following:

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(a) Title, number, and revision of this specification

(b) All drawings and revisions applicable to the procurement item

(c) Data to be submitted or retained (3.2)

(d) When preproduction sample is not required (3.3 and 6.9)

(e) Preproduction sample size, if not as specified (3.3)

(f) Activity designated to perform preproduction evaluation (3.3 and 4.4)

(g) AQL, if not as specified (4.2)

(h) Level of preservation, packaging, and packing required (5.1 and 5.2)

(i) Type of shipping containers to be provided (5.2.1)

(j) Serial numbers and identifying supplier letter assigned (4.2.1)

6.3 Value engineering. Suppliers of items covered by this specification are encouraged to submit value engineering suggestions to the procuring activity in an effort to achieve a required function at the lowest overall cost. Drawings of inert parts are normally "end product" type drawings which permit numerous production methods. If drawings are considered unnecessarily restrictive regarding production methods, a deviation should be requested in time to permit preproduction evaluation.

6.4 Preservation, packaging, and packing. The selection of methods for preservation, packaging, and packing is in accordance with MIL-P-116.

6.5 Security. All parts or assemblies which are classified shall be stored and transported in accordance with appropriate security instructions.

6.6 Deviation. A request for deviation from this specification, applicable drawings, specifications, materials, or processes shall be forwarded to the procuring activity prior to manufacture of the affected item or items. All deviations shall be limited to the contract under which they are granted. A deviation is a before-the-fact request.

6.7 Waiver. A request for waiver from this specification, applicable drawings, specifications, materials, or processes shall be forwarded to the procuring activity. A waiver is an after-the-fact request. A waiver request shall include the following:

(a) Exact nature of the defect

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(b) Characteristic and classification code (symbol and serial number)

(c) Quantity of items involved

(d) Suggested methods of repair.

6.8 Repairs.

6.8.1 Repair by welding. Unless otherwise specified, repair by welding is permitted provided all drawing requirements have been met after repair. All repair by welding shall be accomplished prior to final heat treatment. After final heat treatment, repair by welding shall be made only with prior written approval of the procuring activity.

6.8.2 Repair by plating. Repair by electroplating is not normally permitted but may be requested on an individual part basis for consideration as a waiver.

6.8.3 Thread repair. Thread repair by helical coil is not normally allowed. Helical coil repair in accordance with MS 33537 on such threads will be considered for waiver on an individual part basis.

6.8.4 Repair of defects on parts, welds, etc. Defects found by inspection of parts, welds, forgings, etc, may be removed by grinding, machining, etc, provided that all dimensional and structural requirements are maintained. The repaired area shall be blended or tapered smoothly.

6.8.5 Other repairs. Repairs other than those listed above are not permitted but may be requested on an individual part basis for consideration as a waiver.

6.9 Preproduction sample. A preproduction sample is required from each supplier who has not previously been approved as a qualified supplier of inert parts. If production by a qualified supplier has been interrupted for more than 6 months or changes are requested in production methods which in the opinion of the procuring activity could adversely affect unit characteristics, a preproduction sample may be required.

6.10 Preproduction test activity. The contracting officer should specify the name and location of the activity responsible for preproduction testing (6.2). Unit delivery schedules, location and source of test equipment, and test procedures should be specified.

6.11 Document precedence. Conflicting requirements arising between this specification, the contract and any other applicable documentation shall be referred in writing to the procuring activity for interpretation and clarification.

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